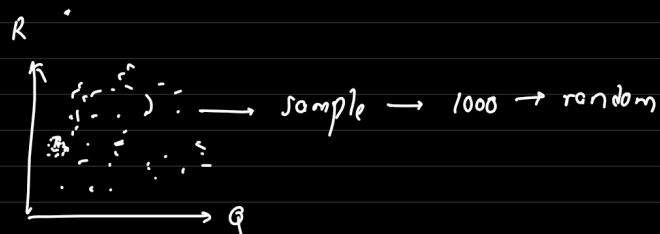


07-03-2026

Agenda:

- * google colab •
- * Data visualization - 2 → $\begin{matrix} \text{matplotlib} \\ \text{seaborn} \end{matrix}$ → better visualization •
- * ~~Database - 1~~



df → random → 1000 → [1, 3, 500, 800,000, 0, 100, 541]

df → random → 1000 → [0, 51, 900, 200 ...]

Seed, random_state → number → 1

[1, 2, 400, 300]

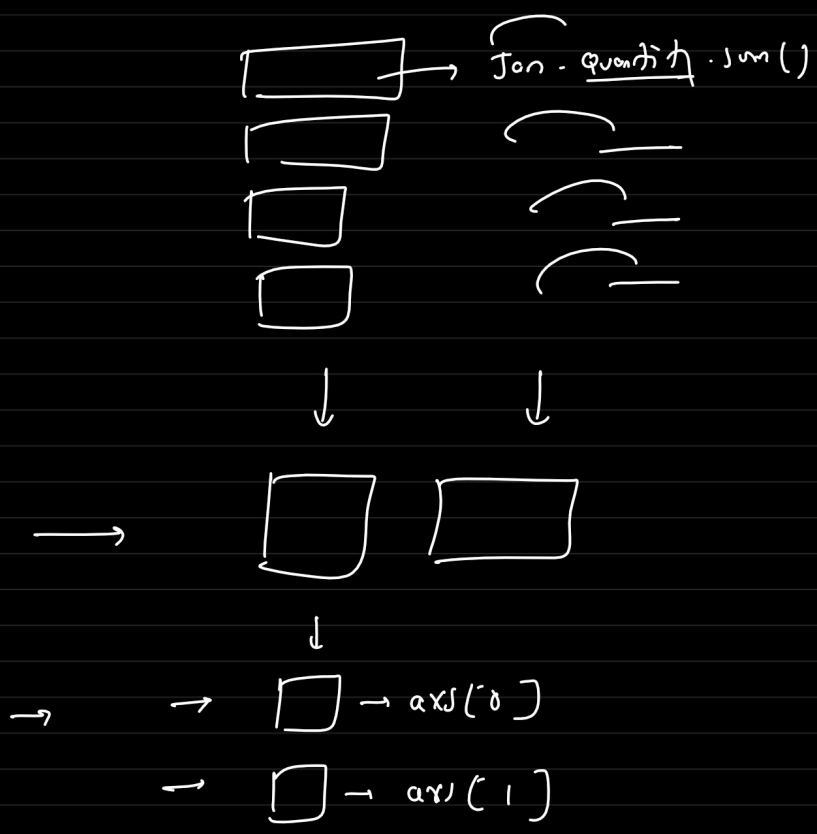
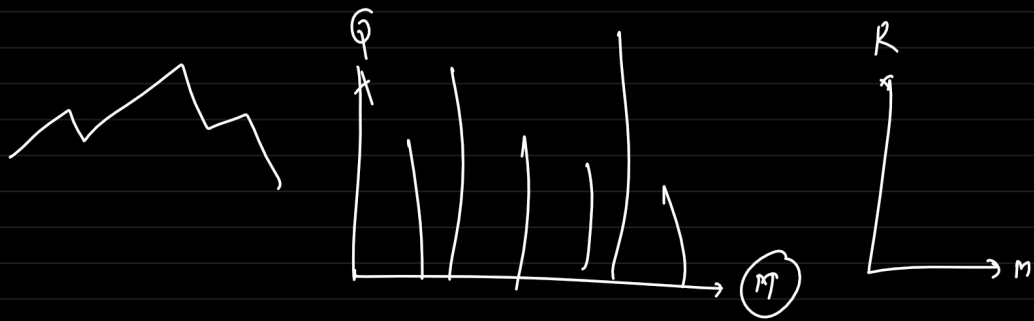
random_state = 1

① → []

① → []

A → 1 → 100 → $\xrightarrow{\text{random_state} = 1}$ [56], A1 → [56]

B → 1-100 → [20]



$nrow = 2$
 $ncol = 2$

0,0	0,1
1,0	1,1

$axis(row_no, col_no)$



Dist of quantity / transaction

How many data is distributed

is it evenly distributed?

→ invoice → quantity → 8
→ 100
→ 200
0 → N

Quantity

invoice → 0 → 100
invoice → 2 → 50
invoice → 100 → 500

0, 1, 2, 3 100

min max

→ 100 points in x-axis



bin → 10 bins

→ 0-10 → C1 81-90
→ 11-20 → C2 91-100 → C10

21-30
31-40
41-50
51-60
61-70
71-80



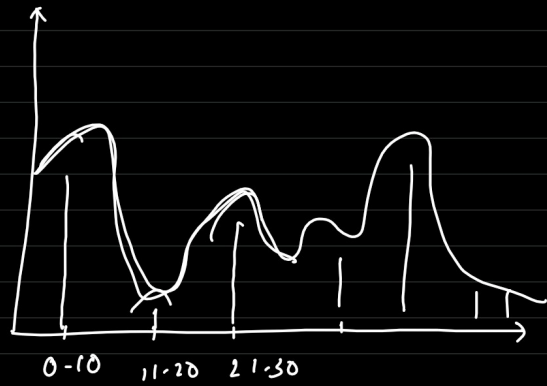
0-10 → total number of quantity

Total number of invoices that have 0-10 quantity

Total number of invoices that have 11-20 quantity

→ 100 invoices → df

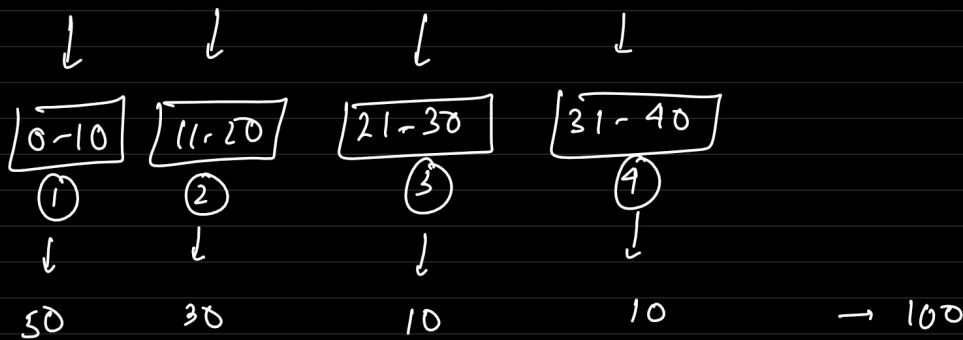
$0 \rightarrow 10 \rightarrow 50 \text{ invoices} \rightarrow 50\%$
 $11 \rightarrow 20 \rightarrow 1$
 $21 \rightarrow 30 \rightarrow 20$
 $31 \rightarrow 40 \rightarrow 20$
 $41 \rightarrow 50 \rightarrow 1$



→ bar

→ density → likelihood of getting
 a invoice with a
 quantity of 0-10

postman



101th time → post man → $1 \rightarrow 50\%$
 $2 \rightarrow 30\%$
 $3 \rightarrow 10\%$
 $4 \rightarrow 10\%$

Kernel Density Estimate

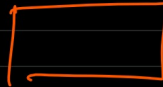
→ Density $\neq$ probability

→ Higher density → High probability



orange →

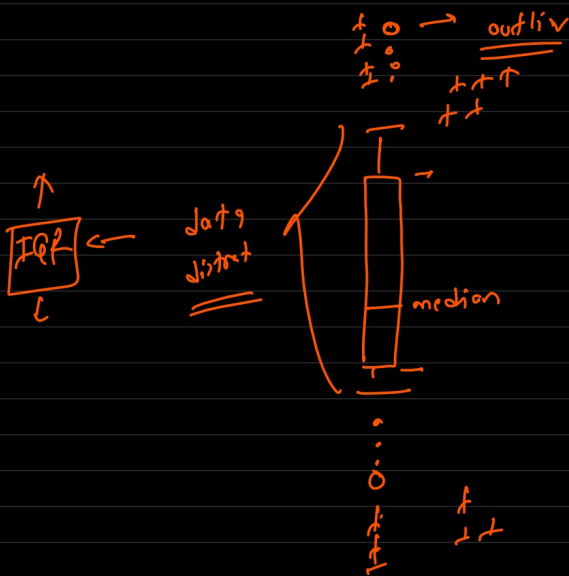
① plt.figure →



② sns.histplot()

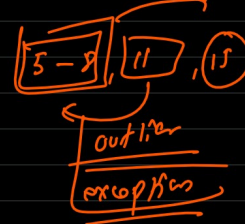
↳ histogram → current active matplotlib context/figure

③ plt.show()



(100) → student → 100

5 → Age →

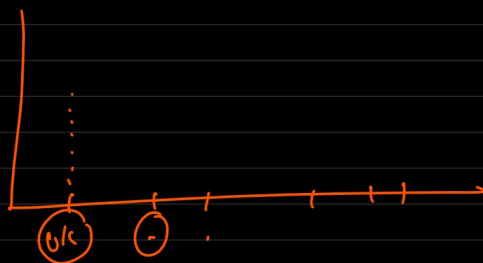


②

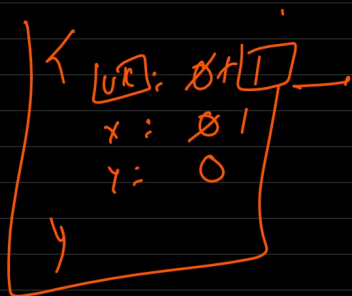
df →

↓ ↓ ↓ ↓

country



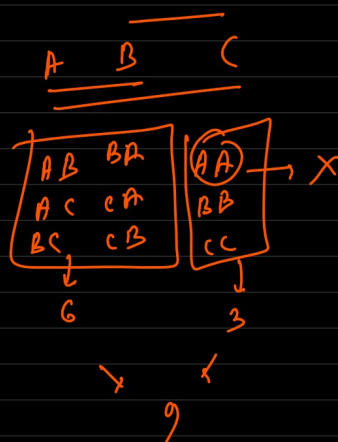
x = df['country']



uk
x
y

100 1000
↑ ↑
x uk y

$$\text{order} = \frac{\text{df}(\text{con}) \cdot \text{value} \cdot \text{sort}}{\text{df}(\text{con}) \cdot \text{value} \cdot \text{sort}}$$



AB
BA

Student	Marks (100)	Rank (100)
	100 ↑	1 ↓
	99	2
	1	100
	5	95

Marks ↑ rank integer ↓

< 0.0

↑ ↓ → inversely proportional → -ve correlation

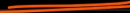
Name Rank

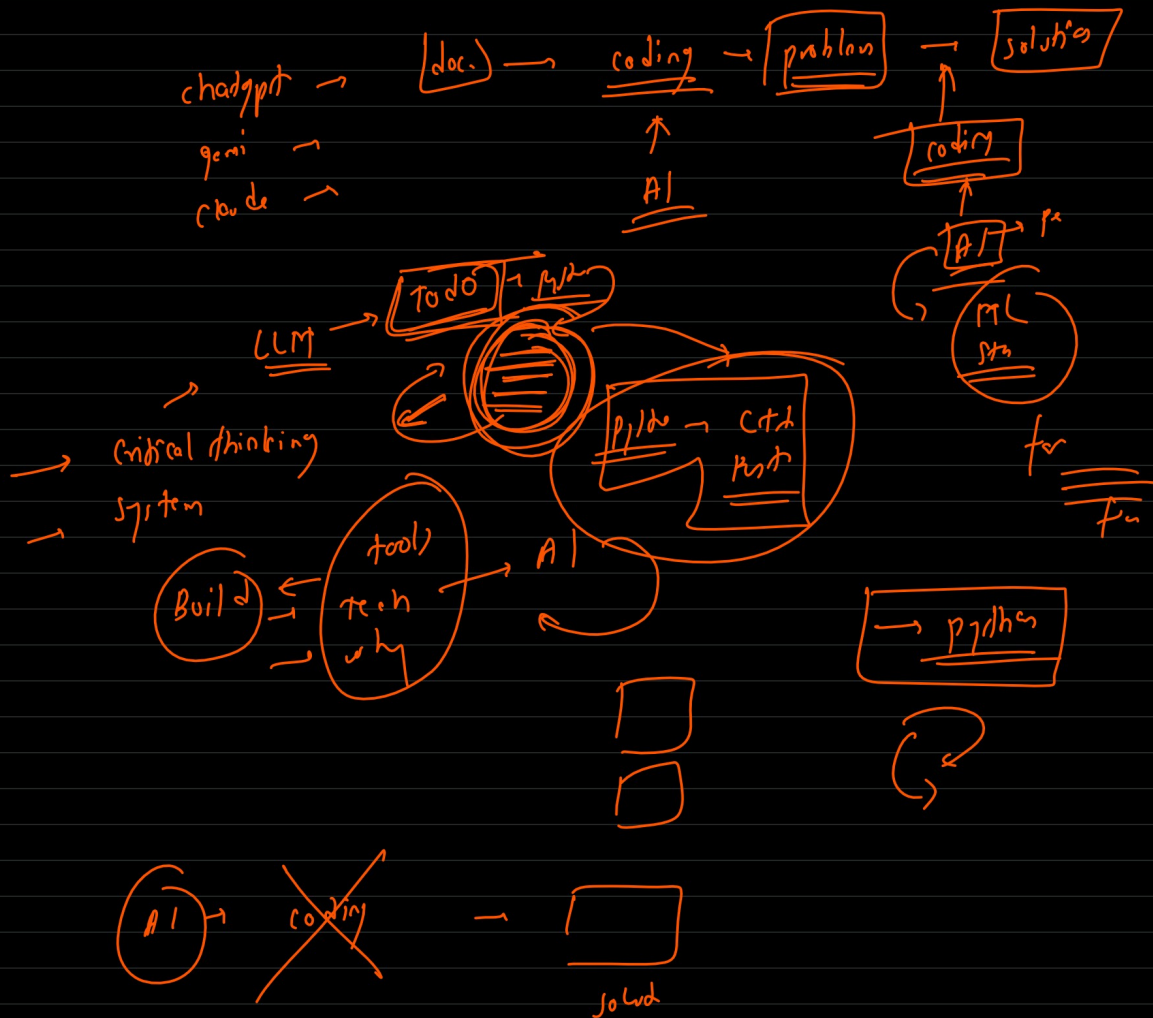
↔ no relation → correlation → 0

money ↑

Investment ↑

directly proportional $\rightarrow$ +ve
 $\rightarrow$ correlation $\rightarrow > 0$

data an → data → plot → 



plm - rock

gut $\rightarrow$ gut h₀

